

ASSIGNMENT-3

1.17. Consider a continuous-time system with input $x(t)$ and output $y(t)$ related by

$$y(t) = x(\sin(t)).$$

- (a) Is this system causal?
- (b) Is this system linear?

1.18. Consider a discrete-time system with input $x[n]$ and output $y[n]$ related by

$$y[n] = \sum_{k=n-n_0}^{n+n_0} x[k],$$

where n_0 is a finite positive integer.

- (a) Is this system linear?
- (a) Is this system time-invariant?
- (c) If $x[n]$ is known to be bounded by a finite integer B (i.e., $|x[n]| < B$ for all n), it can be shown that $y[n]$ is bounded by a finite number C . We conclude that the given system is stable. Express C in terms of B and n_0 .

1.20. A continuous-time linear system S with input $x(t)$ and output $y(t)$ yields the following input-output pairs:

$$\begin{aligned} x(t) = e^{j2t} &\xrightarrow{S} y(t) = e^{j3t}, \\ x(t) = e^{-j2t} &\xrightarrow{S} y(t) = e^{-j3t}. \end{aligned}$$

- (a) If $x_1(t) = \cos(2t)$, determine the corresponding output $y_1(t)$ for system S .
- (b) If $x_2(t) = \cos(2(t - \frac{1}{2}))$, determine the corresponding output $y_2(t)$ for system S .

EXERCISE : What are the properties of the following systems

(a) $y(t) = x(t - 4)$

(c) $y[n] = nx[n]$

(e) $y[n] = \begin{cases} x[n - 1], & n \geq 1 \\ 0, & n = 0 \\ x[n], & n \leq -1 \end{cases}$

(g) $y[n] = x[1 - n]$

(i) $y[n] = \sum_{k=-\infty}^n (\frac{1}{2})^{n-k} x[k]$

(k) $y[n] = \begin{cases} x[n + 1], & n \geq 0 \\ x[n], & n \leq -1 \end{cases}$

(m) $y[n] = x[2n]$

(b) $y(t) = \cos[x(t)]$

(d) $y(t) = \int_{-\infty}^t x(\tau) d\tau$

(f) $y[n] = x[n]x[n - 1]$

(h) $y(t) = \int_{-\infty}^t e^{-(t-\tau)} x(\tau) d\tau$

(j) $y(t) = \frac{dx(t)}{dt}$

(l) $y(t) = x(2t)$

(n) $y[n] = \begin{cases} x[n/2], & n \text{ even} \\ 0, & n \text{ odd} \end{cases}$